Tunable two-species spin models with Rydberg atoms in circular and elliptical states

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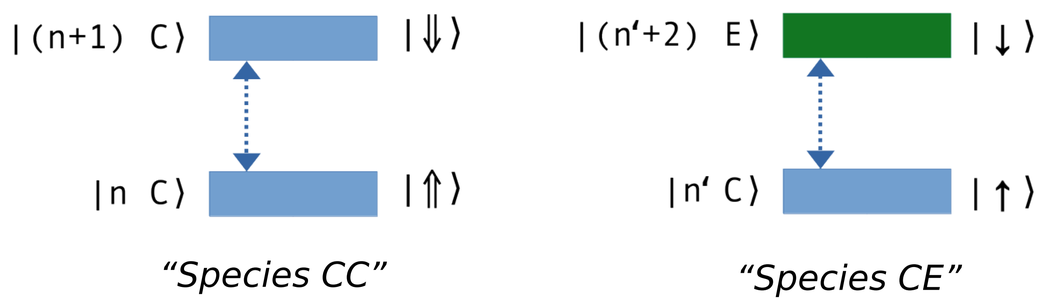
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Abstract

We propose a scheme for creating quantum simulators with Rydberg atoms in long-lived circular and elliptical states. By exciting different subspaces of atomic levels, the atoms can be used to simulate two distinct effective spin species. The spin-spin interaction strength, both intra- and inter-species, can be controlled within a wide range of values. This setup can be used to simulate two-species spin models, or, lattice models with two sublattices.

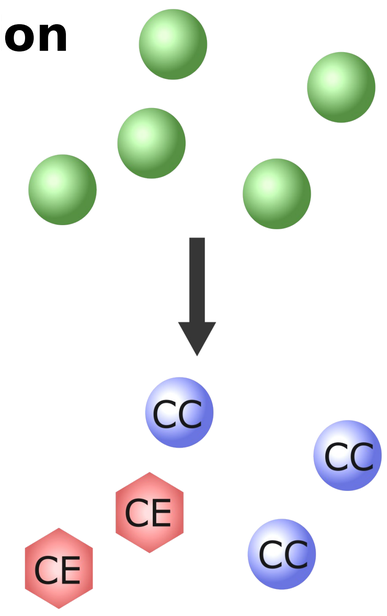
Atoms in circular & elliptical Rydberg states

- **Rydberg atoms:** high quantum number *n*
  - Strong and tunable dipole-dipole interactions; well suited for constructing quantum simulators of interacting particle systems
- **Circular atomic states:** with the maximal magnetic quantum number
$$|m_\ell| = \ell = n\text{-}1$$
- **...and elliptical states:** nearly-maximal magnetic quantum number
$$|m_\ell| = n\text{-}2$$
  - These states have long radiative lifetimes — useful for simulations involving large particle numbers or long times
  - Atoms in these states can be positioned precisely by optical tweezers
- **Using circular & elliptical states to simulate spin models:**
  - Two circular states $|n\,C\rangle$, $|(n+1)\,C\rangle$ — mapped to spin states $|\Uparrow\rangle$, $|\Downarrow\rangle$
  - Dipole-dipole interactions — mapped to spin-spin interactions
  - The transition $|n\,C\rangle \leftrightarrow |(n+1)\,C\rangle$ is dipole-allowed, which lets us obtain effective “spin exchange” interactions
  - Similarly, a pair “circular state + elliptical state” can be mapped to a pair of spin states
  - The different choices of atomic levels result in two different “spin species”, with different spin interaction properties



Two-species spin model simulation

- A two-species spin model can be simulated by a system of atoms excited into the “CC” and “CE” level subspaces
- Interactions between spins can be tuned: (individually) by adjusting distance/angle between any atom pair, and (globally) via electric field



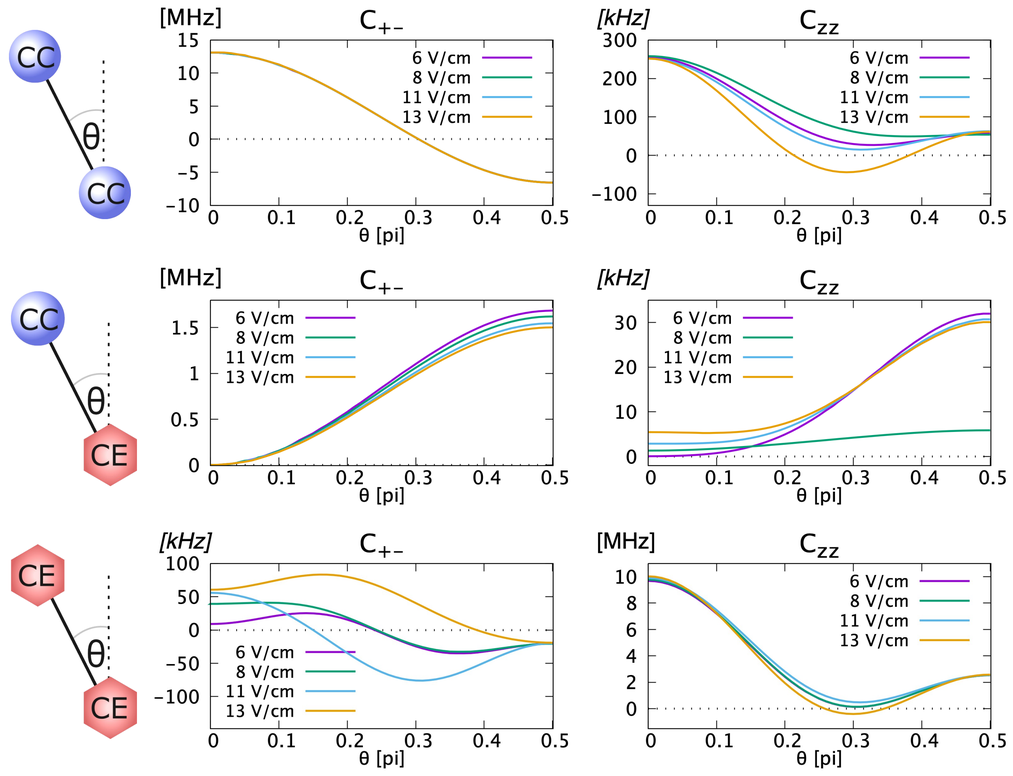
$$\hat{H} = \sum_{j=1}^{N} \hat{h}^{(j)} + \sum_{j<k=1}^{N} \hat{V}^{(j,k)}$$

- **How to obtain the effective spin Hamiltonian H_{eff} *(below)*:**
  - limit the atom Hamiltonian *H* to just our “spin” subspaces
  - calculate the effective interaction for each atom pair; include contributions from first- and higher-order dipolar interactions
  - remove terms corresponding to off-resonant processes

Effective magnetic field | Exchange interaction | Longitudinal interaction

$$\hat{H}_{\text{eff}} = \sum_{j=1}^{N} C_z^j \hat{S}_z^{(j)} + \sum_{j<k=1}^{N} \left(C_{+-}^{jk} \hat{S}_+^{(j)} \hat{S}_-^{(k)} + \text{H.c.}\right) + \sum_{j<k=1}^{N} C_{zz}^{jk} \hat{S}_z^{(j)} \hat{S}_z^{(k)}$$

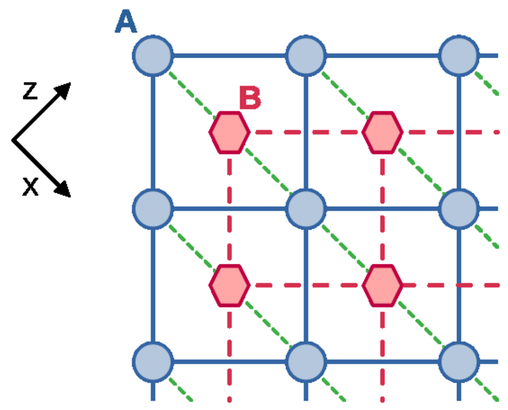
Effective spin interactions – example

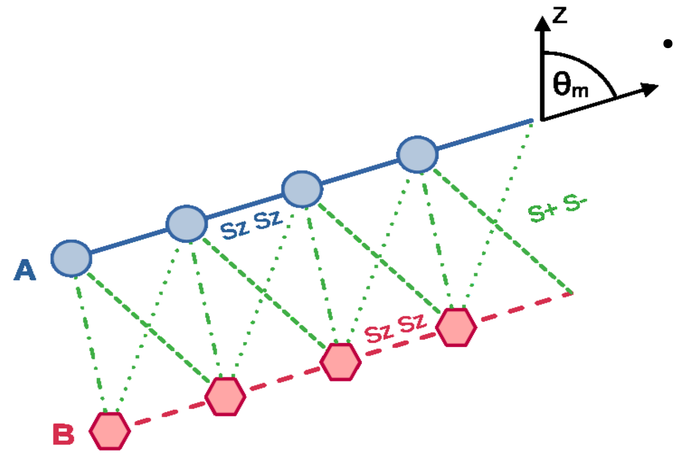


- CC spins: *strong* exchange, *weak* longitudinal; CE spins: *weak* exchange, *strong* longitudinal
- You can tune the interactions via electric field
- Interactions can vanish at certain “magic angles”

Example systems

- **Emulating lattice systems of hardcore bosons:**
  - Atoms = individual sites
  - Internal levels = filling state (site free / occupied)
  - Spin exchange = hopping of bosons between sites
  - Two “species” = two distinct sublattices
- **Dual square lattice**
  - By exploiting magic angles, we can ensure the inter-lattice hopping (green) only occurs along one direction





- **Two coupled Ising chains**
  - Magic angle θm ensures that hopping within each chain disappears, and hopping is only allowed between chains

Read more at: **Phys. Rev. Research 7, 013321**

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